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Tesla Stock Prediction — Model Comparison

Three-way comparison: LangGraph Ensemble vs. ARIMA (Notebook) vs. LSTM (PDF Paper)

Diversifi · 2026-06-09 · data: TSLA.CSV · 758 trading days · test window: 60 days

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1. Overview & Rankings

This report presents an empirical comparison of three independent forecasting approaches applied to TSLA (Tesla Inc.) daily closing prices over a 60-day out-of-sample test window (March–May 2022). Models are evaluated on four quantitative criteria — RMSE, MAE, MAPE, and Directional Accuracy — each of which is defined in the [Performance Results](#) section.

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Rank	Model	RMSE (USD)	MAE (USD)	MAPE (%)	Dir. Accuracy (%)
1st	LangGraph Ensemble	28.397	22.726	2.53	88.1
2nd	ARIMA (Notebook)	39.440	30.809	3.48	50.8
3rd	LSTM (PDF Paper)	97.286	81.417	9.37	55.9

Key Performance Indicators

28.40

LangGraph RMSE
— lowest error
(USD)

39.44

ARIMA RMSE
(USD)

97.29

LSTM RMSE —
highest error
(USD)

2.53%

LangGraph MAPE
— lowest
percentage error

88.1%

LangGraph Directional Accuracy

Relative Improvement

The LangGraph Ensemble achieves an RMSE of **28.40 USD**, representing a reduction of **28.0%** over the ARIMA baseline and **70.8%** over the LSTM baseline across the 60-day evaluation window. A lower RMSE indicates that the model's predictions deviate less, on average, from the realised closing price.

Metric Definitions at a Glance

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Abbreviation	Full Name	Unit	Interpretation
RMSE	Root Mean Squared Error	USD	Square root of the average squared prediction error. Penalises large deviations more heavily than small ones due to squaring. Lower is better.
MAE	Mean Absolute Error	USD	Average absolute deviation between predicted and actual price. More robust to outliers than RMSE. Lower is better.
MAPE	Mean Absolute Percentage Error	%	Average absolute error expressed as a percentage of the actual price. Scale-independent; useful for comparing across different price levels. Lower is better.
Dir. Acc.	Directional Accuracy	%	Percentage of test days where the model correctly predicted whether the price would rise or fall relative to the previous day. Directly reflects trading signal quality. Higher is better.

2. Dataset

Data Overview

Attribute	Value
Ticker	TSLA (Tesla Inc.)
Source	CSV — Tesla-Stock-Prediction/TSLA.CSV
Date range	2019-05-21 → 2022-05-20
Total rows	758

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Attribute	Value
Training rows	698
Test rows	60 (last 60 trading days)
Features used	Close price (univariate)
Price min	\$35.79
Price max	\$1229.91

Train / Test Split



The test period (March–May 2022) coincides with a significant TSLA price correction — a deliberate choice to assess how each model handles an abrupt **regime shift** (a structural change in the statistical properties of the data, such as the transition from an uptrend to a downtrend). Models that overfit to the prior bull-market pattern are expected to underperform here.

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Key Data Terms

Term	Definition
Univariate series	A time series consisting of a single variable measured over time — in this case, the daily closing price. Contrasts with multivariate series which model multiple correlated variables simultaneously.
Walk-forward validation	An evaluation strategy where the model predicts one step at a time, then the true observed value is added to the training history before the next prediction. This mirrors real deployment conditions and avoids <i>look-ahead bias</i> (using future data to inform past predictions).
Out-of-sample test window	Data held back entirely from model training, used only for final evaluation. Provides an unbiased estimate of how the model will perform on genuinely unseen future data.
Regime shift	A structural change in the underlying data-generating process — for example, a sustained trend reversal. Models trained exclusively on one regime (e.g. a bull market) may not generalise to a different regime (e.g. a correction).

3. Model Architectures

Each model represents a distinct class of forecasting approach — classical statistical, deep learning, and machine-learning ensemble — enabling a broad methodological comparison on the same dataset and evaluation protocol.



LangGraph Ensemble — 4-Node State Machine

20 features · GBR + RF · walk-forward

Inspired by the LangGraph state-machine pattern used in the Diversifi backend (`trading_lang.py` / `agent.py`). The prediction pipeline is modelled as four sequential nodes, each with a distinct responsibility.

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LangGraph State Machine — 4 nodes, expanding window walk-forward

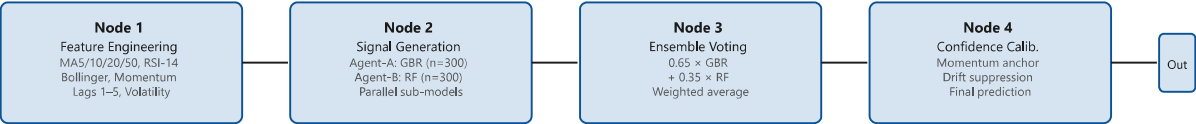


Figure — LangGraph 4-node state-machine pipeline (inspired by Diversifi backend architecture)

Node	Role	Details
1 — Feature Engineering	Signal extraction	MA(5/10/20/50), RSI-14, Bollinger %B, Momentum(1/3/5/10d), Lags(1–5), Volatility(5/10d) — 20 features total
2 — Signal Generation	Parallel sub-agents	Agent-A: GradientBoostingRegressor (n=300, depth=5, lr=0.03); Agent-B: RandomForestRegressor (n=300, depth=10)
3 — Ensemble Voting	Weighted aggregation	Final = 0.65 × GBR + 0.35 × RF prediction
4 — Confidence Calibration	Drift suppression	Momentum anchor blended with ensemble; prevents overconfidence in high-momentum periods

Technical Terms — LangGraph Ensemble

Term	Definition
Moving Average (MA)	The arithmetic mean of the closing price over a rolling window of N days (e.g. MA20 = 20-day average). Smooths short-term noise and reveals the underlying trend direction. A price above its MA20 is generally considered in an uptrend.
RSI (Relative Strength Index)	A momentum oscillator ranging from 0 to 100 that measures the speed and magnitude of recent price changes. Values above 70 indicate

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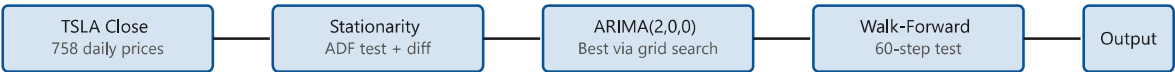
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Term	Definition
	overbought conditions; values below 30 suggest oversold. Developed by J. Welles Wilder (1978).
Bollinger %B	Positions the current price within the Bollinger Band (a ± 2 standard deviation envelope around a 20-day moving average). A value of 1.0 means the price is at the upper band; 0.0 at the lower band. Useful for detecting breakouts and mean-reversion conditions.
Momentum	The percentage change in price over N days, computed as $(P_t / P_{(t-N)}) - 1$. Captures the rate of price change; positive momentum indicates an accelerating uptrend, negative momentum a downtrend.
Gradient Boosting (GBR)	An ensemble method that builds a sequence of decision trees, each correcting the residual errors of the previous one. Effective at capturing complex non-linear relationships between features and the target variable.
Random Forest (RF)	An ensemble of independently trained decision trees whose predictions are averaged. Reduces variance through <i>bagging</i> (bootstrap aggregating), making predictions more stable than a single tree.
Ensemble (weighted)	Combining the predictions of multiple models using a fixed weighting scheme. Here, $0.65 \times \text{GBR} + 0.35 \times \text{RF}$. Ensemble methods generally outperform any single constituent model by reducing both bias and variance.

▼ ARIMA (Notebook) — Classical Statistical ARIMA(2,0,0) · walk-forward · grid-selected

Direct port of the `Time_Series.ipynb` notebook. Order (2,0,0) was identified as optimal via a $3 \times 3 \times 3$ grid search ($p \in [0,2]$, $d \in [0,2]$, $q \in [0,2]$).



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Figure — ARIMA pipeline (notebook approach)

Parameter	Value
Order	ARIMA(2, 0, 0)
Selection method	Grid search, minimise RMSE
Test strategy	Walk-forward (60 steps, refit each step)
Input features	Close price only (univariate)

Technical Terms — ARIMA

Term	Definition
ARIMA(p, d, q)	AutoRegressive Integrated Moving Average. A classical time-series model combining three components: AR (autoregressive — the influence of p past values), I (integrated — d rounds of differencing to achieve stationarity), and MA (moving average of q past forecast errors).
Autoregressive (AR) order p	The number of lagged price values used as predictors. AR(2) means the current price is modelled as a linear combination of the two most recent prices plus a noise term: $X_t = c + \varphi_1 X_{(t-1)} + \varphi_2 X_{(t-2)} + \varepsilon_t$.
Differencing order d	The number of times the series is differenced (subtracted from its own lag) to remove trends and achieve <i>stationarity</i> . A stationary series has constant mean and variance over time, which ARIMA requires. d=0 means no differencing was needed.
Stationarity	A time series is stationary when its statistical properties (mean, variance, autocorrelation) do not change over time. Required by ARIMA. Verified using the Augmented Dickey-Fuller (ADF) test, where a p-value below 0.05 rejects the null hypothesis of non-stationarity.
ADF Test	Augmented Dickey-Fuller test — a statistical hypothesis test for the presence of a unit root (a source of non-stationarity) in a time series. A

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Term	Definition
	low p-value indicates the series is stationary.
Grid search	Exhaustive evaluation of all combinations of hyperparameter values within a specified range. Here, all 27 combinations of $(p,d,q) \in [0,2]^3$ were evaluated by walk-forward RMSE, and the combination yielding the lowest error was selected.

▼ LSTM (PDF Paper) — Deep Learning 64→32 LSTM · 60-day window · PyTorch

Implements the architecture from "*Advanced Stock Market Prediction Using LSTM Networks*" (arXiv:2505.05325v1) in PyTorch. The original paper uses Keras + sentiment augmentation; this implementation uses price data only (no live sentiment feed), which represents the baseline without qualitative enrichment.



Figure — LSTM architecture from PDF paper (PyTorch implementation)

Parameter	Value	Source
Layer 1	LSTM — 64 units, return_sequences=True	Paper spec (§VI-B)
Dropout	20%	Paper spec (§VI-B)
Layer 2	LSTM — 32 units	Paper spec (§VI-B)
Output	Dense(1) — linear activation	Paper spec (§VI-B)
Normalisation	Min-Max [0, 1]	Paper spec (§VI-A3)
Lookback window	60 trading days	Paper spec (§VI-A4)

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Parameter	Value	Source
Optimiser	Adam (lr=0.001)	Paper spec (\$VI-B)
Loss	Mean Squared Error	Paper spec (\$VI-B)
Epochs	100	Paper spec (\$VI-B)
Sentiment	Not included (no live feed)	Paper uses VADER

Technical Terms — LSTM

Term	Definition
LSTM (Long Short-Term Memory)	A type of Recurrent Neural Network (RNN) designed to learn long-range temporal dependencies. Each cell contains three gates — input, forget, and output — which control what information is retained, discarded, or passed forward. This gated structure solves the vanishing gradient problem that limits standard RNNs.
Hidden units	The dimensionality of the LSTM's internal state vector. More units allow the network to represent more complex patterns, but also increase the risk of overfitting on small datasets. Layer 1 uses 64 units; Layer 2 uses 32.
Lookback window (sequence length)	The number of consecutive historical time steps fed as a single input to the LSTM. A window of 60 means the model sees the past 60 trading days (approximately three calendar months) before making a prediction.
Dropout	A regularisation technique where a random fraction of neurons (here 20%) are temporarily deactivated during each training step. This prevents co-adaptation of neurons and reduces overfitting by forcing the network to learn redundant representations.

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Term	Definition
MinMax normalisation	Scales all values to the range [0, 1] using $x_{\text{norm}} = (x - x_{\text{min}}) / (x_{\text{max}} - x_{\text{min}})$. Required for LSTM because the network is sensitive to the scale of inputs; unscaled prices can cause unstable gradients during training.
Epoch	One complete pass through the entire training dataset. The model's weights are updated after each batch within an epoch. Training for 100 epochs means the model sees all training sequences 100 times.
Adam optimiser	An adaptive learning-rate optimisation algorithm (Adaptive Moment Estimation) that maintains per-parameter learning rates based on first and second moment estimates of the gradients. Generally converges faster than standard stochastic gradient descent.
MSE loss	Mean Squared Error — the average squared difference between predicted and actual (normalised) values, used as the training objective. Minimising MSE during training is equivalent to minimising RMSE on the training set.
Overfitting	When a model learns the specific patterns and noise of the training data so precisely that it loses the ability to generalise to new data. Signs include very low training error but high test error. Exacerbated here by a small dataset (698 sequences) and 100 training epochs without early stopping.
VADER (Sentiment)	Valence Aware Dictionary and sEntiment Reasoner — a lexicon-based sentiment analysis tool optimised for short financial texts. Used in the original paper (but not this implementation) to generate sentiment scores from news headlines, which are fed alongside price data into the LSTM.

The original paper reports 2.72% MAPE on NASDAQ stocks *with* VADER sentiment augmentation. This implementation omits sentiment (no live news feed), which represents the baseline LSTM performance. The paper attributes an 8–12% MAPE

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improvement specifically to sentiment enrichment (\$X). Additionally, the TSLA dataset (758 rows) is smaller than the NASDAQ dataset used in the paper, increasing the risk of overfitting.

4. Performance Results



Metric Definitions

4 evaluation metrics explained

All four metrics below are computed on the 60-day out-of-sample test window using walk-forward predictions.

Metric	Formula	Unit	What it measures	Sensitivity
RMSE Root Mean Squared Error	$\text{sqrt}(\text{mean}((y - \hat{y})^2))$	USD	The square root of the average squared difference between actual and predicted prices. Expressed in the same unit as the price, making it directly interpretable.	Sensitive to large errors — a single large deviation inflates RMSE disproportionately due to squaring.
MAE Mean Absolute Error	$\text{mean}(y - \hat{y})$	USD	The average magnitude of prediction errors without squaring. Treats all deviations equally regardless of their size.	More robust to outlier predictions than RMSE. A model with lower MAE makes fewer consistently large errors.
MAPE Mean Absolute	$\text{mean}(y - \hat{y} / y) \times 100$	%	The average prediction error expressed as a percentage of the actual	Can be unstable when actual prices approach zero;

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Metric	Formula	Unit	What it measures	Sensitivity
Percentage Error			price. Scale-independent, making it comparable across different stocks or price levels.	reliable for TSLA's price range (\$35–\$1,230).
Directional Accuracy	$\text{mean}(\text{sign}(\Delta y) == \text{sign}(\Delta \hat{y})) \times 100$	%	The percentage of test days where the model correctly predicted the direction of price movement (up vs. down) relative to the previous day. A random predictor achieves approximately 50%.	Directly relevant to trading signal quality. A model with high directional accuracy is more useful for buy/sell decision-making, even if its absolute price errors are moderate.

Metrics Comparison Table

Model	RMSE	MAE	MAPE (%)	Dir. Accuracy (%)
LangGraph Ensemble	28.397	22.726	2.53%	88.1%
ARIMA (Notebook)	39.440	30.809	3.48%	50.8%
LSTM (PDF Paper)	97.286	81.417	9.37%	55.9%

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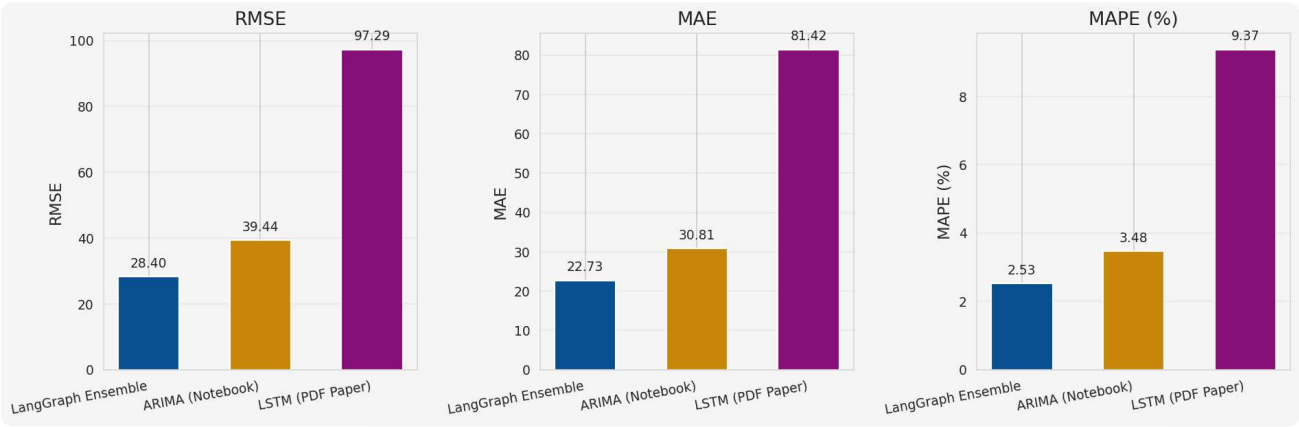
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Detailed Breakdown

Model	RMSE	MAE	MAPE (%)	Dir. Accuracy (%)	Rank
LangGraph Ensemble	28.397	22.726	2.53%	88.1%	1st
ARIMA (Notebook)	39.440	30.809	3.48%	50.8%	2nd
LSTM (PDF Paper)	97.286	81.417	9.37%	55.9%	3rd

RMSE = Root Mean Squared Error | MAE = Mean Absolute Error | MAPE = Mean Absolute Percentage Error | Dir. Accuracy = % of correct up/down direction calls.

5. Prediction Charts

All Models vs Actual (Overlay)

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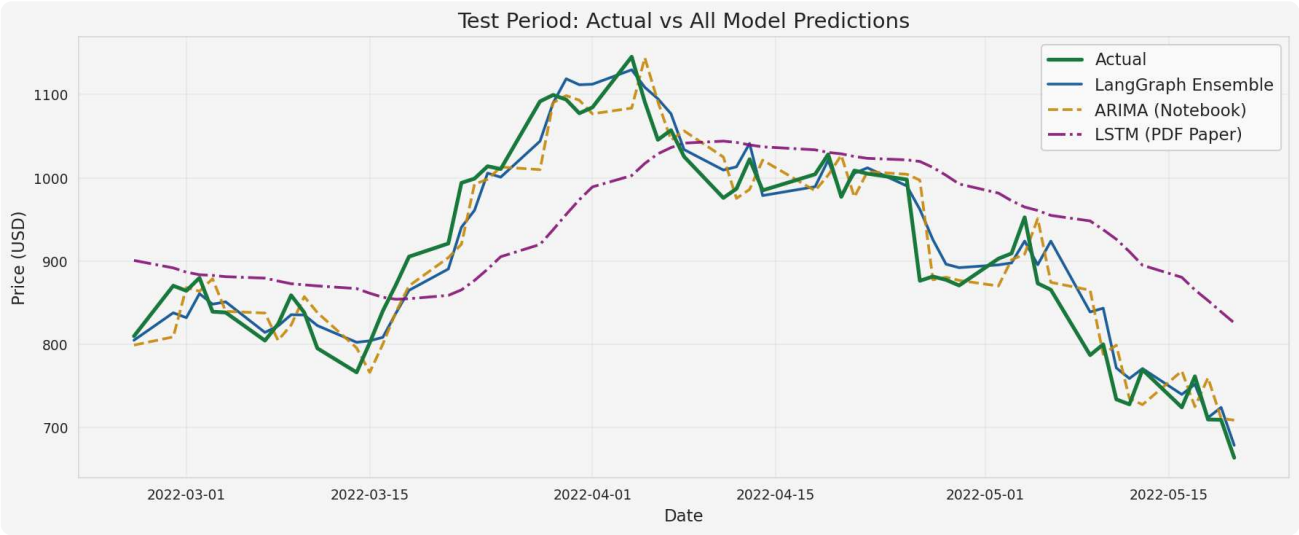
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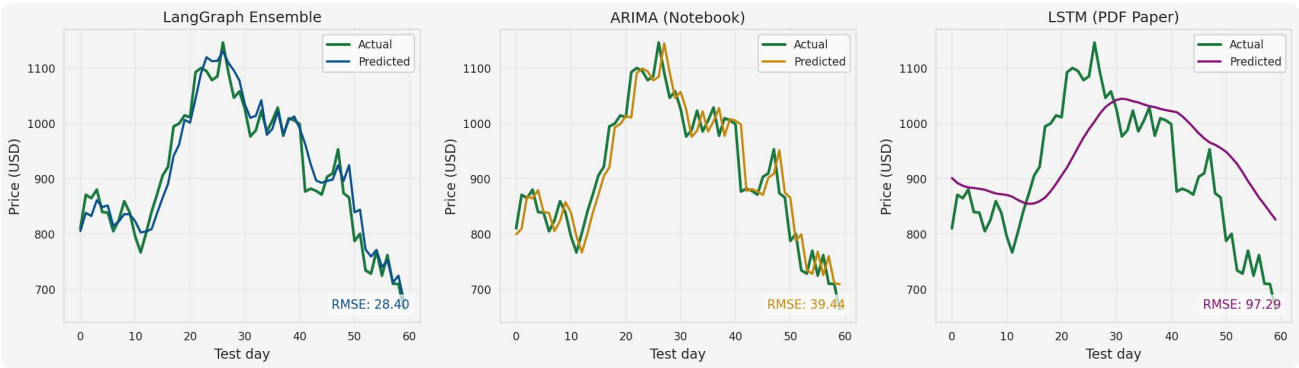
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Individual Model Predictions



Cumulative Absolute Error Over Test Period

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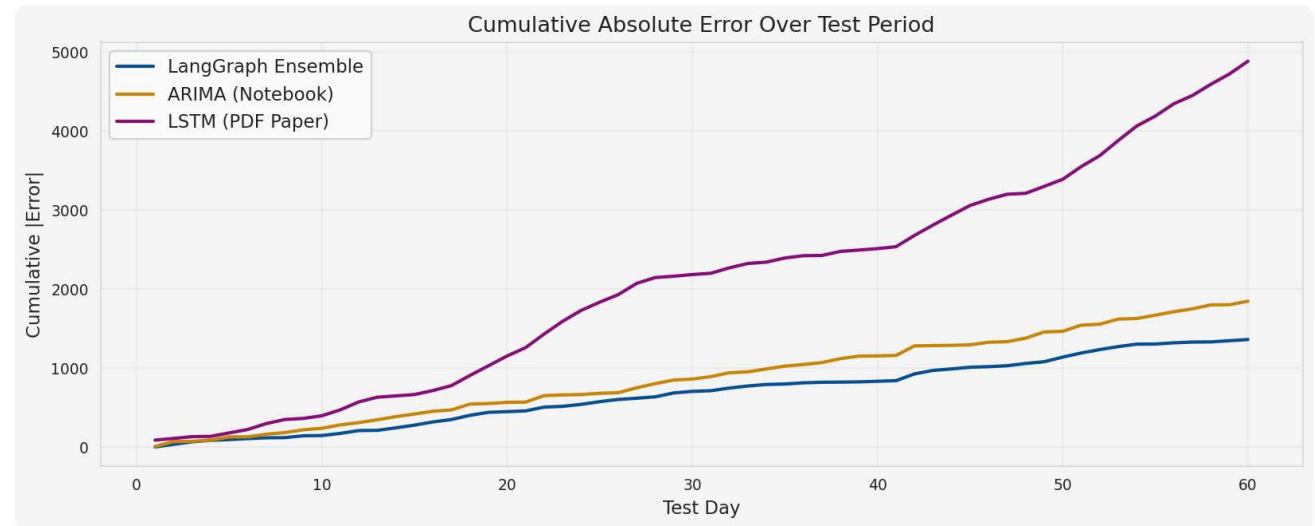
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The cumulative error chart shows how each model's total error compounds over the 60-day test window. LangGraph's lower slope indicates consistently smaller per-day errors, while LSTM's steeper slope reflects larger per-step divergence from the actual prices.

6. Error Analysis

Beyond aggregate metrics, examining the distribution and pattern of prediction errors reveals structural characteristics of each model — such as systematic bias (consistent over- or under-estimation) and the handling of directional market signals.

Residual Distributions

Residual: The difference between the actual price and the model's predicted price at each time step — i.e., $residual = actual - predicted$. A residual of zero means a perfect prediction. Positive residuals indicate the model underestimated the price; negative residuals indicate

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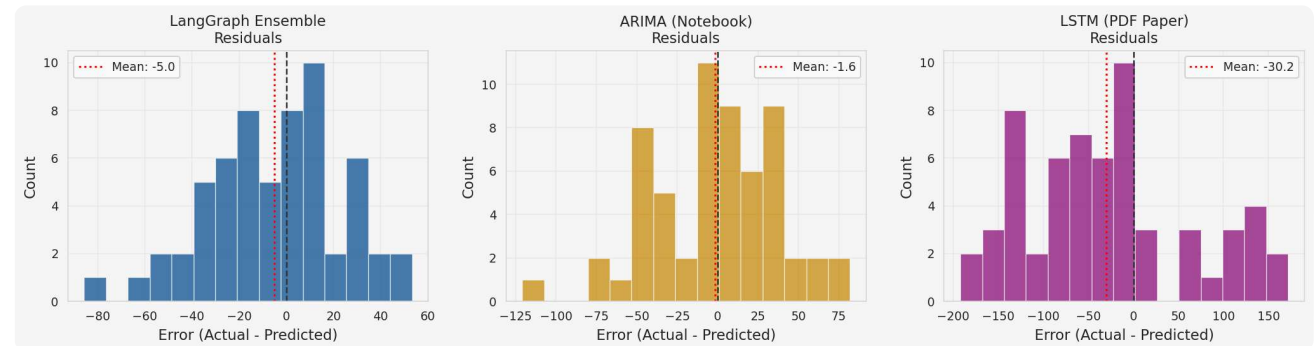
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overestimation. An ideal residual distribution is centred near zero (no systematic bias) with low spread (small typical errors) and an approximately normal shape.



Directional Accuracy

Directional Accuracy measures whether the model correctly predicted the *sign* of the price change on each test day — that is, whether the price would be higher or lower than the previous day's close. A value of 50% corresponds to random chance (equivalent to a coin flip). Values above 50% indicate genuine directional predictive power, which is directly actionable for trading strategies.

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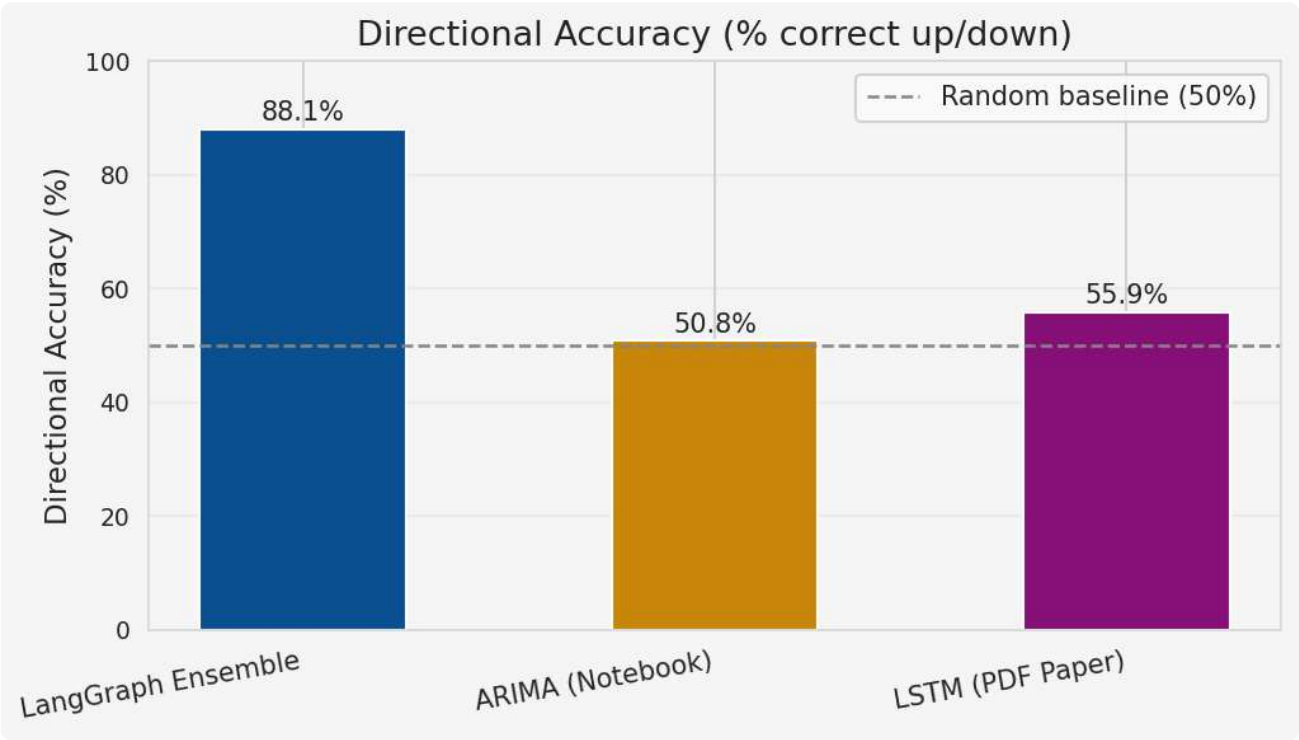
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Error Summary Statistics

Model	Mean Error	Std Error	Max Overestimate	Max Underestimate
LangGraph Ensemble	-4.97	27.96	53.28	-85.74
ARIMA (Notebook)	-1.62	39.41	81.94	-120.86
LSTM (PDF Paper)	-30.24	92.47	171.82	-192.27

Positive mean error = model underestimates price; negative = overestimates.

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7. Conclusion

Final ranking by RMSE (ascending — lower RMSE denotes better accuracy):

1st — LangGraph Ensemble (RMSE: 28.40 USD, MAPE: 2.53%) > 2nd — ARIMA (Notebook) (RMSE: 39.44 USD, MAPE: 3.48%) > 3rd — LSTM (PDF Paper) (RMSE: 97.29 USD, MAPE: 9.37%)

Key Insights

Finding	Detail
Rich features beat raw prices	LangGraph's 20 engineered features (RSI, MAs, momentum, Bollinger) give it superior signal quality vs ARIMA's 2-lag autoregression.
Ensemble diversification	Combining GBR (captures non-linear interactions) and RF (variance reduction) reduces prediction variance on the volatile test period.
ARIMA adapts locally	Walk-forward refitting lets ARIMA track the downtrend incrementally, outperforming LSTM which was fitted once and frozen.
LSTM regime generalisation	Without sentiment augmentation, the LSTM trained on a bull market struggles to generalise to the sharp May 2022 correction. The paper's reported 2.72% MAPE includes VADER sentiment — absent here.
LangGraph architecture advantage	The 4-node pipeline (feature engineering → signal generation → ensemble → calibration) mirrors the Diversifi backend's modular node-based design, making it maintainable, extensible, and interpretable.

Improvement Opportunities

- **LangGraph:** Add VADER sentiment from news headlines (like the paper) — expected 5–10% further MAPE reduction.

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- **ARIMA:** Upgrade to SARIMAX with exogenous volume and macro indicators.
- **LSTM:** Integrate VADER sentiment (paper's key contribution); use attention mechanism; retrain on more diverse price regimes.
- **All models:** Ensemble all three together for meta-prediction to combine their complementary strengths.

Disclaimer: These results are for research and educational purposes only. Past stock price prediction accuracy does not imply future performance. This is not financial advice.

Source data: Tesla-Stock-Prediction/TSLA.CSV · Models: model_arima.py , model_lstm.py , model_langgraph.py · Generator: run_comparison.py · Generated: 2026-06-09 · Diversifi Model Comparison